

DDL COMMANDS, CONSTRAINTS

CREATING DATA BASE:

A new database named *employee_db* was created using SQL with the following statement:

```
1      -- create data base
2 •    create database Employee_db;
3 •    use Employee_db;
```

CREATING TABLES:

A table named *department*, *location*, *employee* was created in the database to store details, using the following SQL statement.

```
4 • CREATE TABLE Departments (
5     department_id INT PRIMARY KEY,
6     department_name VARCHAR(100)
7 );
8 • CREATE TABLE Location (
9     location_id INT PRIMARY KEY,
10    location_name VARCHAR(100)
11 );
12 • CREATE TABLE Employees (
13     employee_id INT PRIMARY KEY,
14     employee_name VARCHAR(100),
15     gender VARCHAR(10) ,
16     age INT ,
17     hire_date date ,
18     designation VARCHAR(100),
19     salary DECIMAL(10,2),
20     department_id INT,
21     location_id INT,
22     FOREIGN KEY (department_id)
23         REFERENCES Departments(department_id)
24         ON DELETE CASCADE
25         ON UPDATE CASCADE,
26     FOREIGN KEY (location_id)
27         REFERENCES Location(location_id)
28         ON DELETE CASCADE
29         ON UPDATE CASCADE
30 );
```

DDL COMMENTS:

The DDL comments included in the SQL file describe the structure and constraints of the database objects.

- `ALTER TABLE Employees
ADD email VARCHAR(100);`
- `ALTER TABLE Employees
MODIFY designation VARCHAR(200);`
- `ALTER TABLE Employees
DROP COLUMN age;`
- `ALTER TABLE Employees
rename column hire_date to date_of_joining ;`
- `RENAME TABLE Departments TO Departments_Info;`
- `RENAME TABLE Location TO Locations;`
- `TRUNCATE TABLE Employees;`
- `DROP TABLE employees;`
- `DROP DATABASE employee_db;`

CONSTRAINTS:

The database was recreated with the specified constraints to ensure proper structure and data integrity.

```
-- Create the database
CREATE DATABASE employee_db;
USE employee_db;

-- 1. Departments Table
CREATE TABLE Departments_Info (
    department_id INT PRIMARY KEY,
    department_name VARCHAR(100) NOT NULL UNIQUE
);

-- 2. Locations Table
CREATE TABLE Locations (
    location_id INT auto_increment primary KEY,
    location_name VARCHAR(100) NOT NULL UNIQUE
);

-- 3. Employees Table
CREATE TABLE Employees (
    employee_id INT PRIMARY KEY,
    employee_name VARCHAR(100) NOT NULL,
    gender VARCHAR(10) CHECK (gender IN ('Male', 'Female')),
    age INT CHECK (age >= 18),
    hire_date DATE NOT NULL,
    designation VARCHAR(200),
    salary DECIMAL(10,2),
    department_id INT,
    location_id INT,
    email VARCHAR(100) UNIQUE,
    FOREIGN KEY (department_id)
        REFERENCES Departments_Info(department_id)
        ON DELETE CASCADE
        ON UPDATE CASCADE,
    FOREIGN KEY (location_id)
        REFERENCES Locations(location_id)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);

ALTER TABLE Employees
MODIFY hire_date DATE DEFAULT (CURRENT_DATE);
```

INSERT VALUES TO DATABASE:

Values were inserted into the tables in accordance with the defined constraints.

```
-- Insert into Departments_Info
```

```
INSERT INTO Departments_Info (department_id, department_name)
```

VALUES (1, 'Human Resources'),

(2, 'Finance'),

(3, 'IT'),

```
(4, 'Marketing');
```

```
-- Insert into Locations
```

```
INSERT INTO Locations (location_id, location_name)
```

VALUES (1, 'Chennai'),

```
(2, 'Bengaluru'),
```

(3, 'Delhi'),

```
(4, 'Mumbai');
```

```
-- Insert into Employees
```

```
INSERT INTO Employees (employee id, employee name, gender, age, hire date, designation, salary, department id, location_id, email)
```

VALUES

```
(101, 'Arun Kumar', 'Male', 28, DEFAULT, 'Data Analyst', 45000.00, 3, 2, 'arun.kumar@example.com'),
```

```
(102, 'Priya Sharma', 'Female', 32, DEFAULT, 'HR Manager', 60000.00, 1, 1, 'priya.sharma@example.com'),
```

```
(103, 'Ravi Patel', 'Male', 26, DEFAULT, 'Software Engineer', 52000.00, 3, 4, 'ravi.patel@example.com'),
```

```
(104, 'Meena Iyer', 'Female', 29, DEFAULT, 'Marketing Executive', 40000.00, 4, 3, 'meena.iyer@example.com');
```

```
select *from employees;
```

Data entries were added to the table according to the defined schema and constraints.

[illegible]

ENHANCED ENTITY–RELATIONSHIP DIAGRAM:

This image shows the relationships between the tables in the database schema.

